Yujun Zhou

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Summary

I am a Ph.D. student at the University of Notre Dame, advised by Prof. Xiangliang Zhang, graduating in **Spring 2027**. My research includes LLM reasoning and reinforcement learning, long-horizon coding agents, and reliable evaluation and alignment. Previously, I interned at Microsoft and Tencent AI Lab.

Seeking U.S. full-time research roles starting Spring/Summer 2027.

Education

2023 - Present	PhD in Computer Science at **University of Notre Dame, USA**	(GPA: 4.00/4.0)
2021 - 2022	Master's Degree in Computer Science at **KAUST, Saudi Arabia**	(GPA: 3.83/4.0)
2017 - 2021	Bachelor's Degree in Computer Science at **UESTC, China**	(GPA: 3.92/4.0)

Selected Publications

* Equal contribution.

[1] **Yujun Zhou**, Jingdong Yang, Yue Huang, Kehan Guo, Zoe Emory, Bikram Ghosh, Amita Bedar, Sujay Shekar, Zhenwen Liang, Pin-Yu Chen, Tian Gao, Werner Geyer, Nuno Moniz, Nitesh V. Chawla, and Xiangliang Zhang. "Benchmarking large language models on safety risks in scientific laboratories". ***Nature Machine Intelligence*** 8(1), 20–31 (2026). doi:10.1038/s42256-025-01152-1.

[2] Chuan Zou, **Yujun Zhou**, Xiaonan Luo, and Xiangliang Zhang. "StateHarness: Maintaining Evolving Project State for Long-Horizon Coding Agents". *arXiv preprint*, 2026. arXiv:2610.17616.

[3] **Yujun Zhou**, Kehan Guo, Haomin Zhuang, Xiangqi Wang, Yue Huang, Zhenwen Liang, Pin-Yu Chen, Tian Gao, Nuno Moniz, Nitesh V. Chawla, Xiangliang Zhang. "Getting Better at Working With You: Compiling User Corrections into Runtime Enforcement for Coding Agents". In *arXiv preprint*. arXiv:2606.13174. [Code]

[4] **Yujun Zhou***, Yue Huang*, Han Bao*, Kehan Guo, Zhenwen Liang, Pin-Yu Chen, Tian Gao, Werner Geyer, Nuno Moniz, Nitesh V Chawla, Xiangliang Zhang. "Alignment Risks from Capability-Seeking RL Training". In: ***ICML***. 2026.

[5] **Yujun Zhou***, Zhenwen Liang*, Haolin Liu, Wenhao Yu, Kishan Panaganti, Linfeng Song, Dian Yu, Xiangliang Zhang, Haitao Mi, Dong Yu. "Evolving Language Models without Labels: Majority Drives Selection, Novelty Promotes Variation". In *arXiv preprint*. arXiv:2509.15194.

[6] **Yujun Zhou***, Jiayi Ye*, Zipeng Ling*, Yufei Han, Yue Huang, Haomin Zhuang, Zhenwen Liang, Kehan Guo, Taicheng Guo, Xiangqi Wang, and Xiangliang Zhang. "Dissecting Logical Reasoning in LLMs: A Fine-Grained Evaluation and Supervision Study". In: ***EMNLP Findings***. 2025.

[7] **Yujun Zhou**, Yufei Han, Haomin Zhuang, Taicheng Guo, Kehan Guo, Zhenwen Liang, Hongyan Bao, and Xiangliang Zhang. "Defending Jailbreak Prompts via In-Context Adversarial Game". In: ***EMNLP***. 2024.

[8] **Yujun Zhou***, Yufei Han*, Haomin Zhuang, Hongyan Bao, and Xiangliang Zhang. "Attack-free Evaluating and Enhancing Adversarial Robustness on Categorical Data". In: ***ICML***. 2024.

[9] Zhenwen Liang*, **Yujun Zhou***, Sidi Lu, Xiangliang Zhang, Haitao Mi, Dong Yu. "Too Correct to Learn: Reinforcement Learning on Saturated Reasoning Data". In **ACL**. 2026.

[10] Zhenwen Liang*, Ruosen Li*, **Yujun Zhou***, Linfeng Song, Dian Yu, Xinya Du, Haitao Mi, Dong Yu. "Your Reasoning Model is Secretly a Reward Model - Optimization-Free Verification from Experience". In **ACL**. 2026.

[11] Xiangqi Wang, Yue Huang, Yanbo Wang, Xiaonan Luo, Kehan Guo, **Yujun Zhou**, Xiangliang Zhang. "AdaReasoner: Adaptive Reasoning Enables More Flexible Thinking". In ***NeurIPS***. 2025 (Spotlight).

Research Experience

LLM Reasoning and Reinforcement Learning

- **EVOL-RL: Label-Free Reasoning RL.** Developed an evolution-inspired framework combining majority-based selection with novelty rewards to mitigate entropy collapse, preserve reasoning diversity, and improve generalization without labeled rewards.
- **FineLogic: Logical Reasoning Evaluation and Supervision.** Evaluated accuracy, stepwise soundness, and representation alignment; compared four supervision formats to study their generalization to logical reasoning.

Long-Horizon Coding Agents

- **StateHarness: Evolving Project State.** Studied runtime support for maintaining project state across long-horizon coding tasks, including findings, decisions, repository relationships, and unfinished work, without updating model parameters.
- **TRACE: Runtime Enforcement of User Corrections.** Developed a skill-layer pipeline that compiles user corrections into runtime checks before task completion. On ClawArena's in-distribution held-out evaluation, preference violations fell from 100% to 37.6% while preserving task success.

Trustworthy Evaluation and Alignment

- **Alignment Risks from Capability-Seeking RL Training.** Investigated how capability-seeking RL can induce exploitation of training-time vulnerabilities and deceptive behaviors that mimic genuine performance gains.
- **LabSafety Bench.** Developed a benchmark for evaluating LLM reliability and safety awareness in scientific laboratories, extending evaluation to safety-critical real-world scenarios.
- **In-Context Adversarial Game.** Studied LLM-agent-based adversarial learning for defending against jailbreak prompts without model fine-tuning.

Industry Experience

Microsoft — Research Intern May–Aug. 2026

- **Enterprise AI Benchmarking and Trustworthy LLM Evaluation.** Built realistic enterprise-oriented benchmarks for LLM agents, focusing on evidence-aware reasoning, reliable decision-making, and trustworthy behavior in complex, long-horizon workplace scenarios.

Tencent AI Lab (Tencent America LLC) — Research Intern May–Dec. 2025

- **LLM Reasoning and Reinforcement Learning.** Researched label-free RL and mathematical reasoning in the AGI (Audio/Speech/Language & Multimodal) team, including EVOL-RL, reasoning diversity, and methods for expanding problem-solving capabilities.

Services

Workshop Organizer: RelSciFM Workshop, KDD 2026.
Invited Conference/Journal Reviewer: ICLR, NeurIPS, ARR, COLM, AAAI, KDD, CVPR, ICML